João Natividade

João da Cruz de Natividade e Silva Neto (João Silva Neto) · Belém, Brazil · Remote

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PROFESSIONAL SUMMARY

Software engineer and Computer Engineering senior at UFPA with strong hands-on experience in backend development, microservices, MCP automation, and cloud computing (AWS). Practical expertise maintaining and optimizing Python APIs (FastAPI) and Rust systems, resolving memory leaks in cross-platform apps (Flutter/Dart), building interactive dashboards with Dash, and developing high-performance browser graphics (120 FPS Canvas).

PROFESSIONAL EXPERIENCE

Software Engineering Intern · LaPS / UFPA (DMóvel Assistive Technology)

March 2026 — September 2026 | Belém, Brazil | 360 Hours

- Maintained and refactored Python/FastAPI backend integrated with databases and cloud services on Amazon Web Services (AWS).
- Developed and maintained cross-platform assistive UI in Flutter/Dart and interactive dashboards using Dash (Python), HTML, CSS, and JavaScript.
- Diagnosed and resolved memory leaks and runtime performance bottlenecks in the production mobile app.
- Managed Git branch workflows and produced technical demonstration materials for the SITEC 2026 conference.
- Graded 10.0/10 (Concept Excellent) by the permanent lab advisor.

SELECTED PROJECTS

AetherSense (Rust, ESP32, React, Python, ONNX)

An experimental platform connecting Wi-Fi signals, processing and a web interface to explore camera-free sensing.

- Link: joaosnet.github.io/en/projects/aethersense/ | Código: github.com/joaosnet/aethersense

DMóvel (Flutter, Dart, FastAPI, AWS, Dash)

Assistive technology ecosystem featuring a Flutter mobile app, an async FastAPI backend on AWS, and analytical telemetry dashboards.

- Link: lapshub.github.io/dmovel_web_app/ | Código: github.com/LAPSHub/dmovel_web_app

Varedura (Python, AsyncIO, Docker, MCP, Segurança)

Fast CLI tool and MCP connector for subdomain enumeration, network probing, and attack-surface discovery.

- Link: joaosnet.github.io/en/projects/varedura/ | Código: github.com/joaosnet/Varedura

Evolutionary Lab (JavaScript, Plotly, AG, PSO, ED)

A browser-based lab comparing genetic algorithms, particle swarm optimization and differential evolution on benchmark functions.

- Link: joaosnet.github.io/evolutionary-optimization-viz-ag-pso/ | Código: github.com/joaosnet/evolutionary-optimization-viz-ag-pso

TECHNICAL SKILLS

Languages: Python, Rust, JavaScript, TypeScript, Dart, C, SQL, HTML5, CSS3

Backend & Cloud: FastAPI, AWS, Docker, REST APIs, MCP (Model Context Protocol), Dash

Systems & Tools: Git, GitHub Actions, Linux, ESP32, Canvas 2D, Three.js, WebSockets

EDUCATION

Bachelor of Science in Computer Engineering — Universidade Federal do Pará (UFPA)

Bachelor's degree in progress · Expected graduation: 2026 | Belém, Brazil

- Academic Honors: Maximum grade (100/100) in Hardware Design, Software Engineering, Databases, and Cloud Computing.
- Languages: Portuguese (Native) | English (Advanced)